

KOLYVAN', Russian Federation

WMO: 296310

Lat: **55.322N**Lon: **82.739E**Elev: **482**StdP: **14.44**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.9	-26.2	-36.6	0.7	-31.6	-31.1	1.0	-25.9	27.7	13.3	23.4	17.5	4.4	40	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	84.3	67.1	81.4	66.0	78.6	64.7	70.0	79.7	68.2	77.6	66.5	75.6	6.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.7	100.1	74.5	64.8	93.7	72.7	63.1	88.1	71.2	34.4	79.8	32.8	77.5	31.5	75.5	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	19.2	16.5	DB	-34.3	89.7	7.6	3.2	-39.8	92.0	-44.2	93.9	-48.5	95.7	-54.1	98.0
			WB	-34.4	72.6	7.6	2.2	-39.9	74.2	-44.3	75.5	-48.6	76.7	-54.1	78.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.4	-0.1	6.2	21.0	38.7	52.9	63.3	66.3	61.9	50.5	37.2	19.3	6.0	
	DBStd	25.59	15.28	13.18	11.18	10.26	10.13	8.02	5.81	6.59	8.34	9.31	13.53	14.95	
	HDD50	6924	1553	1226	898	362	90	7	0	2	95	407	921	1363	
	HDD65	11029	2018	1646	1363	790	392	124	55	139	442	861	1371	1828	
	CDD50	1604	0	0	0	22	179	406	506	370	109	12	0	0	
	CDD65	234	0	0	0	0	17	73	97	42	5	0	0	0	
	CDH74	1907	0	0	0	6	215	612	677	343	54	0	0	0	
	CDH80	401	0	0	0	1	52	132	137	70	9	0	0	0	
Wind	WSAvg	6.9	7.8	7.7	7.7	7.3	7.1	5.7	4.9	5.1	6.1	7.1	7.9	8.3	
Precipitation	PrecAvg	18.5	0.9	0.7	0.8	1.1	1.3	2.2	2.8	2.3	1.6	1.7	1.5	1.4	
	PrecMax	26.1	2.0	1.5	1.7	2.0	3.0	5.0	4.8	5.4	3.6	3.2	2.8	2.4	
	PrecMin	13.0	0.2	0.1	0.1	0.2	0.0	0.7	0.4	0.4	0.6	0.5	0.9	0.6	
	PrecStd	3.3	0.5	0.4	0.4	0.5	0.8	1.1	1.2	1.1	0.7	0.7	0.6	0.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	34.5	46.2	72.7	85.6	87.3	87.8	86.3	80.4	66.5	46.4	34.2	
		MCWB	31.5	32.1	38.8	54.5	62.7	67.5	70.6	68.0	62.9	52.9	43.3	32.6	
	2%	DB	28.2	31.0	41.6	65.4	79.6	83.7	83.7	81.5	73.6	60.0	42.2	32.0	
		MCWB	26.6	29.1	35.9	51.8	60.4	65.9	69.1	66.5	59.5	49.7	39.2	31.1	
	5%	DB	25.1	27.9	38.1	60.0	75.1	80.7	80.8	77.8	68.7	54.9	38.7	29.1	
		MCWB	23.7	26.1	34.0	48.3	57.9	64.9	67.7	64.7	57.2	47.3	36.5	27.8	
	10%	DB	21.5	24.6	35.3	54.5	70.5	77.4	77.8	74.2	64.0	50.2	35.1	25.6	
		MCWB	20.1	23.1	32.1	45.7	55.9	63.7	66.2	63.0	54.9	44.7	33.3	24.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.8	32.7	40.1	56.2	65.5	70.3	73.5	71.5	64.2	54.0	44.2	33.0
			MCDB	33.5	34.0	45.6	68.5	82.3	82.6	83.8	79.7	76.0	63.5	46.0	33.5
2%		WB	26.8	29.2	36.6	52.9	62.1	68.3	71.1	68.2	60.6	50.9	39.6	31.1	
		MCDB	28.0	30.8	40.3	64.2	76.9	79.6	80.4	77.5	71.2	58.2	41.4	32.0	
5%		WB	23.8	26.1	34.3	49.3	59.2	66.6	69.2	66.1	58.2	48.1	36.4	27.9	
		MCDB	25.1	27.7	37.5	58.7	72.3	77.4	78.0	75.2	67.3	54.2	38.6	29.0	
10%		WB	20.2	23.1	32.4	45.6	56.7	64.9	67.5	64.2	55.6	45.1	33.4	24.2	
		MCDB	21.4	24.6	35.3	54.1	68.5	75.1	75.6	72.5	62.5	49.6	35.0	25.4	
Mean Daily Temperature Range	5% DB	MDBR	12.3	13.5	14.7	16.7	20.0	18.2	17.1	18.0	17.2	12.7	10.6	12.0	
		MCDBR	13.3	12.7	15.1	24.8	25.8	22.3	20.5	22.9	23.9	21.1	10.1	11.9	
		MCWBR	12.9	11.8	11.2	14.3	12.4	9.6	9.0	10.3	11.8	12.3	8.3	11.8	
	5% WB	MCDBR	13.0	12.4	13.5	23.1	23.7	20.0	18.2	20.5	22.4	19.5	10.0	12.1	
		MCWBR	12.9	11.8	10.8	14.3	12.8	9.8	9.3	10.5	11.8	12.0	8.7	12.2	
Clear-Sky Solar Irradiance	taub	0.291	0.325	0.373	0.407	0.404	0.407	0.436	0.409	0.370	0.352	0.322	0.293		
	taud	2.006	1.967	1.916	2.081	2.220	2.290	2.208	2.270	2.327	2.242	2.055	1.954		
	Ebn at Noon	185	224	242	256	266	266	254	254	248	218	170	148		
	Edh at Noon	22	34	45	45	41	39	42	37	30	26	21	19		
All-Sky Solar Radiation	RadAvg	289	613	1084	1466	1678	1889	1695	1417	952	480	233	182		
	RadStd	26	34	69	102	178	179	114	105	115	57	41	29		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	-1.10	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A